JOWORINGO SIMBARASHE

Mechanical Engineering Undergraduate

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EDUCATION

Bachelor of Technology in Mechanical Engineering
Jawaharlal Nehru Technological University, Kakinada (JNTUK)
Expected Graduation: May 2027 | CGPA: 8.84/10

PROJECTS

UAV System Design & Integration

- Designed system-level CAD models for a fixed-wing UAV fuselage and wings using SolidWorks, optimizing for specific mission payload and range requirements.
- Executed design segmentation strategy for additive manufacturing to reduce part count while maintaining structural integrity and assembly alignment.
- Planned complex internal wiring routes for avionics and control surfaces, minimizing electromagnetic interference and service access time.
- Performed conceptual aircraft sizing and aerodynamic trade-off analysis (wing loading, aspect ratio, lift–drag behavior) for low-Reynolds-number flight
- Analyzed aerodynamic trade-offs between wing aspect ratio and structural stiffness, selecting a high-lift airfoil for short takeoff capabilities.

Supercharger Effects on IC Engine Performance Analysis

- Developed Python scripts to analyze and visualize supercharger impact on engine performance, creating comprehensive performance curves for various boost pressures.
- Modeled thermodynamic parameters including air-fuel ratio, combustion efficiency, and volumetric efficiency, achieving a remarkable 134% volumetric efficiency.
- Analyzed the influence of intake pressure and temperature rise on engine breathing characteristics and overall performance.
- Generated and interpreted performance maps and characteristic curves (boost pressure vs power, efficiency vs speed) using Matplotlib and Pandas.

CERTIFICATIONS

Python for Mechanical Engineers	*Decibels Lab*
CATIA - Perform as a Mechanical Designer	*Coursera*
SOLIDWORKS 3D CAD for Education	*Coursera*
Modeling and Simulation with Simulink	*Coursera*
Introduction to Programming with MATLAB	*Coursera*
Learning Fusion 360	*Autodesk*
Intro to Siemens NX: Engineering Essentials	*Coursera*

ENGINEERING INTERESTS

- **Applied Automation:** Designing robust systems for resource-constrained environments.
- **Energy Systems:** HVAC efficiency and thermal management optimization.

TECHNICAL SKILLS

Programming

Python: Engineering analysis, Data visualization (Matplotlib/Seaborn)

MATLAB: Algorithm development, Control system design

Analysis & Simulation

- Simulink & Simscape Modeling
- Computational Fluid Dynamics (CFD)
- Finite Element Analysis (FEA)
- Kinematic System Modeling

TOOLS & SOFTWARE

SolidWorks | Autodesk Fusion 360 | CATIA | Siemens NX | ANSYS Fluent | MATLAB/Simulink | Python (NumPy/Pandas) | MS Office Suite | 3D Printing (FDM)

LANGUAGES

English (Professional Working Proficiency)
Shona (Native)